

Exercise 1

Source Code: rigidbody2d_collision.py

Starting Point: rigidbody2d_collision_hack.py

Additions I Added:

- Square-square collision using the Separating Axis Theorem
- Full relative velocity at the contact point (angular and linear velocity accounted for)
- Momentum impulse w/ rotational effects

Exercise 2

a) Coefficient of Restitution = 1.0

Scenario	Collision(s)
0	impulse=LVector3f(-4.3636364, 0, 0) time=2.5167 contact_point=LVector3f(-0.016665183, 25, 0) normal=LVector3f(1, 0, 0) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.066658943, 25, -0.5), LVecBase3f(-0.06665897, 25, 0.5), LVecBase3f(0.033328592, 25, 0.5), LVecBase3f(0.033328592, 25, -0.5)] averaged_contact=LVector3f(-0.016665183, 25, 0) corner_from=body0, face_from=body1
1	impulse=LVector3f(-4.2459707, 0, 0) time=2.4500 contact_point=LVector3f(0.2220122, 25, -0.15938297) normal=LVector3f(1, 0, 0) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(0.20000791, 25, -0.09586203), LVecBase3f(0.26602077, 25, -0.13397461), LVecBase3f(0.20000791, 25, -0.24831223)] averaged_contact=LVector3f(0.2220122, 25, -0.15938297) corner_from=body0, face_from=body1
2	impulse=LVector3f(-4.0509595, 0, -2.3388223) time=2.4833 contact_point=LVector3f(-0.26289227, 25, 0.13397455) normal=LVector3f(0.86602544, 0, 0.5) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.2993511, 25, 0.13397455), LVecBase3f(-0.27200692, 25, 0.18133586), LVecBase3f(-0.21731879, 25, 0.08661322)]

	averaged_contact=LVector3f(-0.26289227, 25, 0.13397455) corner_from=body1, face_from=body0
3	impulse=LVector3f(0, 0, -0.805396) time=1.7000 contact_point=LVector3f(-2.4036233, 25, 0.9001664) normal=LVector3f(0, 0, 1) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-2.4031517, 25, 0.9), LVecBase3f(-2.40419, 25, 0.9), LVecBase3f(-2.403528, 25, 0.9004991)] averaged_contact=LVector3f(-2.4036233, 25, 0.9001664) corner_from=body0, face_from=body1
4	impulse=LVector3f(-2.465567, 0, -0.7734598) time=2.5167 contact_point=LVector3f(0.12118425, 25, -0.63634973) normal=LVector3f(0.95415216, 0, 0.29932194) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(0.12218156, 25, -0.65845584), LVecBase3f(0.11075023, 25, -0.59362554), LVecBase3f(0.13062097, 25, -0.65696775)] averaged_contact=LVector3f(0.12118425, 25, -0.63634973) corner_from=body1, face_from=body0 impulse=LVector3f(-3.2704732, 0, -0.9518215) time=2.5500 contact_point=LVector3f(-0.13653694, 25, 0.5002886) normal=LVector3f(0.96016293, 0, 0.27944082) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.18143558, 25, 0.6018516), LVecBase3f(-0.13729214, 25, 0.60830223), LVecBase3f(-0.09088309, 25, 0.29071184)] averaged_contact=LVector3f(-0.13653694, 25, 0.5002886) corner_from=body0, face_from=body1

b) Coefficient of Restitution = 0.0

Scenario	Collision(s)
0	Because the blocks stick together, an absurd number of collisions register, and I can't reasonably paste all the output here, a total of 57 collisions registered.
1	Similar to scenario 0, 27 collisions were registered.

2	impulse=LVector3f(-2.0254797, 0, -1.1694112) time=2.4833 contact_point=LVector3f(-0.26289227, 25, 0.13397455) normal=LVector3f(0.86602544, 0, 0.5) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.2993511, 25, 0.13397455), LVecBase3f(-0.27200692, 25, 0.18133586), LVecBase3f(-0.21731879, 25, 0.08661322)] averaged_contact=LVector3f(-0.26289227, 25, 0.13397455) corner_from=body1, face_from=body0
3	impulse=LVector3f(0, 0, -0.402698) time=1.7000 contact_point=LVector3f(-2.4036233, 25, 0.9001664) normal=LVector3f(0, 0, 1) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-2.4031517, 25, 0.9), LVecBase3f(-2.40419, 25, 0.9), LVecBase3f(-2.403528, 25, 0.9004991)] averaged_contact=LVector3f(-2.4036233, 25, 0.9001664) corner_from=body0, face_from=body1
4	impulse=LVector3f(-1.2327835, 0, -0.3867299) time=2.5167 contact_point=LVector3f(0.12118425, 25, -0.63634973) normal=LVector3f(0.95415216, 0, 0.29932194) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(0.12218156, 25, -0.65845584), LVecBase3f(0.11075023, 25, -0.59362554), LVecBase3f(0.13062097, 25, -0.65696775)] averaged_contact=LVector3f(0.12118425, 25, -0.63634973) corner_from=body1, face_from=body0 impulse=LVector3f(-0.1366016, 0, -0.033321827) time=2.5333 contact_point=LVector3f(0.09600614, 25, -0.6022239) normal=LVector3f(0.9715132, 0, 0.2369855) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(0.10783159, 25, -0.67361915), LVecBase3f(0.06436816, 25, -0.4610666), LVecBase3f(0.115818634, 25, -0.6719859)] averaged_contact=LVector3f(0.09600614, 25, -0.6022239) corner_from=body1, face_from=body0 impulse=LVector3f(-1.480888, 0, -0.34986466) time=2.5500 contact_point=LVector3f(-0.034951906, 25, -0.026212617) normal=LVector3f(0.97320866, 0, 0.22992375) contact_type=multi-contact (averaged)

	<pre> raw_contacts=[LVecBase3f(0.10009236, 25, -0.689686), LVecBase3f(-0.21166146, 25, 0.6298888), LVecBase3f(-0.1308338, 25, 0.6440415), LVecBase3f(0.102595284, 25, -0.6890947)] averaged_contact=LVector3f(-0.034951906, 25, -0.026212617) corner_from=body0, face_from=body1 impulse=LVector3f(-0.6665259, 0, -0.18776653) time=2.5667 contact_point=LVector3f(-0.093948, 25, 0.2742755) normal=LVector3f(0.9625356, 0, 0.27115523) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.22618127, 25, 0.56700754), LVecBase3f(-0.0816831, 25, 0.58406633), LVecBase3f(0.026020385, 25, -0.32824742)] averaged_contact=LVector3f(-0.093948, 25, 0.2742755) corner_from=body0, face_from=body1 impulse=LVector3f(-0.00016587619, 0, -0.000054431813) time=2.5833 contact_point=LVector3f(-0.09707005, 25, 0.29547548) normal=LVector3f(0.9501512, 0, 0.3117895) contact_type=multi-contact (averaged) raw_contacts=[LVecBase3f(-0.2270391, 25, 0.5140109), LVecBase3f(-0.056251347, 25, 0.52615183), LVecBase3f(-0.007919688, 25, -0.15373635)] averaged_contact=LVector3f(-0.09707005, 25, 0.29547548) corner_from=body0, face_from=body1 </pre>
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Exercise 3

a) Coefficient of Restitution = 1.0

Scenario	Initial and Final Metrics
0	Initial: <pre> body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 </pre>

total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000
total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000

Final:

body0_pos=LPoint3f(-18.615266, 25, -0.5) body0_vel=LVector3f(-2.3636364, 0, 0) body0_omega=-3.272727
body0_momentum=LVector3f(-2.3636364, 0, 0)
body0_ke_bulk=2.793389 body0_ke_spin=3.570248
body0_L_bulk_y=1.181818 body0_L_spin_y=-2.181818
body1_pos=LPoint3f(3.6484901, 25, 0.5) body1_vel=LVector3f(0.3636365, 0, 0)
body1_omega=-3.272727
body1_momentum=LVector3f(0.3636365, 0, 0)
body1_ke_bulk=0.066116 body1_ke_spin=3.570248
body1_L_bulk_y=0.181818 body1_L_spin_y=-2.181818
total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=2.859505 total_ke_spin=7.140496 total_ke=10.000001
total_L_bulk_y=1.363636 total_L_spin_y=-4.363636 total_L_y=-3.000000

Comments:

Angular Momentum

Before:

Ly=-3.000000 (all bulk, no spin)

After:

Ly=-3.000000

- Lbulk,y=1.363636
- Lspin,y=-4.363636

Conclusion: Total angular momentum is conserved. The collision redistributes angular momentum from bulk (orbital) motion into spin, but the total remains unchanged.

Energy

Before:

- KEbulk=10.000000
- KEspin=0.000000
- Total E=10.000000

After:

- KEbulk=2.859505
- KEspin=7.140496
- Total E=10.000001≈10E

	Conclusion: Total kinetic energy is conserved (within numerical precision). Energy is transformed from translational kinetic energy into rotational (spin) kinetic energy due to a tangential impulse during the collision. No energy is lost from the system; it is simply redistributed between different forms of mechanical motion. Conservation Summary • Total linear momentum: conserved • Total angular momentum: conserved • Total energy: conserved • Individual components (bulk vs spin energy and angular momentum): not conserved separately, but redistributed during the collision.
1	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000 total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000 Final: body0_pos=LPoint3f(-18.019638, 25, -0.5) body0_vel=LVector3f(-2.2459707, 0, 0) body0_omega=-2.169375 body0_momentum=LVector3f(-2.2459707, 0, 0) body0_ke_bulk=2.522192 body0_ke_spin=1.568729 body0_L_bulk_y=1.122985 body0_L_spin_y=-1.446250 body1_pos=LPoint3f(3.0530071, 25, 0.5) body1_vel=LVector3f(0.24597072, 0, 0) body1_omega=-4.199581 body1_momentum=LVector3f(0.24597072, 0, 0) body1_ke_bulk=0.030251 body1_ke_spin=5.878827 body1_L_bulk_y=0.122985 body1_L_spin_y=-2.799721 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=2.552443 total_ke_spin=7.447556 total_ke=10.000000 total_L_bulk_y=1.245971 total_L_spin_y=-4.245971 total_L_y=-3.000000 Comments:

	Angular Momentum Before: Ly=-3.000000 (all bulk, no spin) After: Ly=-3.000000 • Lbulk,y=1.245971 • Lspin,y=-4.245971 Conclusion: Total angular momentum is conserved. The collision redistributes angular momentum from bulk (orbital) motion into spin, while keeping the total unchanged. Energy Before: • KEbulk=10.000000 • KEs spin=0.000000 • Total E=10.000000 After: • KEbulk=2.552443 • KEs pin=7.447556 • Total E=10.000000 Conclusion: Total kinetic energy is conserved. Energy is transformed from translational kinetic energy into rotational (spin) kinetic energy during the collision (bulk KE drops by 7.447557 while spin KE increases by 7.447556, up to rounding). Conservation Summary • Total linear momentum: conserved • Total angular momentum: conserved • Total energy: conserved • Individual components (bulk vs spin energy and angular momentum): not conserved separately, but redistributed during the collision.
2	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000

body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000
body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0)
body1_omega=0.000000
body1_momentum=LVector3f(-4, 0, 0)
body1_ke_bulk=8.000000 body1_ke_spin=0.000000
body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000
total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000
total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000

Final:

body0_pos=LPoint3f(-16.4155, 25, -18.041212)
body0_vel=LVector3f(-2.0509595, 0, -2.3388223) body0_omega=-1.149404
body0_momentum=LVector3f(-2.0509595, 0, -2.3388223)
body0_ke_bulk=4.838263 body0_ke_spin=0.440377
body0_L_bulk_y=-1.391144 body0_L_spin_y=-0.766269
body1_pos=LPoint3f(1.4488892, 25, 18.041212)
body1_vel=LVector3f(0.050959587, 0, 2.3388223) body1_omega=2.440299
body1_momentum=LVector3f(0.050959587, 0, 2.3388223)
body1_ke_bulk=2.736343 body1_ke_spin=1.985020
body1_L_bulk_y=-2.469322 body1_L_spin_y=1.626866
total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=7.574606 total_ke_spin=2.425396 total_ke=10.000002
total_L_bulk_y=-3.860466 total_L_spin_y=0.860597 total_L_y=-2.999869

Comments:

Angular Momentum

Before:

Ly=-3.000000 (all bulk, no spin)

After:

Ly=-2.999869≈-3

- Lbulk,y=-3.860466
- Lspin,y=0.860597

Conclusion: Total angular momentum is conserved to numerical precision. The collision redistributes angular momentum between bulk and spin (bulk becomes more negative while spin becomes positive), with the total essentially unchanged.

Energy

Before:

- KEbulk=10.000000
- KEspin=0.000000

	• Total E=10.000000E = 10.000000E=10.000000 After: • KEbulk=7.574606 • KEspin=2.425396 • Total E=10.000002≈10 Conclusion: Total kinetic energy is conserved to numerical precision. Energy is transformed from translational kinetic energy into rotational (spin) kinetic energy (bulk KE decreases by ≈2.425394 while spin KE increases by ≈2.425396, up to rounding). Also, the bulk kinetic energy is redistributed across components (motion is no longer purely along x; significant z-motion appears after the collision). Conservation Summary • Total linear momentum: conserved; total momentum remains (-2,0,0) • Total angular momentum: conserved to numerical precision • Total energy: conserved to numerical precision • Individual components (bulk vs spin energy and angular momentum): not conserved separately, but redistributed during the collision.
3	Initial: <pre>body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=-5.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=8.333333 body0_L_bulk_y=-1.000000 body0_L_spin_y=-3.333333 body1_pos=LPoint3f(-1.9, 25, 1.9) body1_vel=LVector3f(0, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(0, 0, 0) body1_ke_bulk=0.000000 body1_ke_spin=0.000000 body1_L_bulk_y=0.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(2, 0, 0) total_ke_bulk=2.000000 total_ke_spin=8.333333 total_ke=10.333333 total_L_bulk_y=-1.000000 total_L_spin_y=-3.333333 total_L_y=-4.333333</pre> Final: <pre>body0_pos=LPoint3f(13.966736, 25, -7.1714) body0_vel=LVector3f(2, 0, -0.805396) body0_omega=-4.762755 body0_momentum=LVector3f(2, 0, -0.805396) body0_ke_bulk=2.324331 body0_ke_spin=7.561277 body0_L_bulk_y=-3.094047 body0_L_spin_y=-3.175170 body1_pos=LPoint3f(-1.9, 25, 8.571394) body1_vel=LVector3f(0, 0, 0.805396) body1_omega=0.608424 body1_momentum=LVector3f(0, 0, 0.805396) body1_ke_bulk=0.324331 body1_ke_spin=0.123393</pre>

body1_L_bulk_y=1.530252 body1_L_spin_y=0.405616
total_momentum=LVector3f(2, 0, 0)
total_ke_bulk=2.648663 total_ke_spin=7.684671 total_ke=10.333333
total_L_bulk_y=-1.563794 total_L_spin_y=-2.769553 total_L_y=-4.333348

Comments:

Angular Momentum

Before:

Ly=-4.333333

- Lbulk,y=-1.000000
- Lspin,y=-3.333333

After:

Ly=-4.333348≈-4.333333

- Lbulk,y=-1.563794
- Lspin,y=-2.769553

Conclusion: Total angular momentum is conserved to numerical precision ($\Delta L_y \approx -1.5 \times 10^{-5}$). Angular momentum is transferred **from spin to bulk** overall (spin becomes less negative, bulk becomes more negative), with redistribution between the two bodies.

Energy

Before:

- KEbulk=2.000000
- KEspin=8.333333
- Total E=10.333333

After:

- KEbulk=2.648663
- KEspin=7.684671
- Total E=10.333333

Conclusion: Total kinetic energy is conserved. Energy is transformed from **rotational (spin) kinetic energy** into **translational kinetic energy** (spin KE decreases by ≈ 0.648662 while bulk KE increases by ≈ 0.648663 , up to rounding). This is consistent with the collision converting some initial spin into linear motion (notably generating z-velocity for both bodies).

Conservation Summary

- **Total linear momentum:** conserved; total momentum remains (2,0,0)
- **Total angular momentum:** conserved to numerical precision

	• Total energy: conserved • Individual components (bulk vs spin energy and angular momentum): not conserved separately, but redistributed during the collision.
4	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=3.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=3.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=2.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=10.000000 total_ke_spin=3.000000 total_ke=13.000000 total_L_bulk_y=-3.000000 total_L_spin_y=2.000000 total_L_y=-1.000000 Final: body0_pos=LPoint3f(-28.753408, 25, -13.350294) body0_vel=LVector3f(-3.7360403, 0, -1.7252812) body0_omega=0.940150 body0_momentum=LVector3f(-3.7360403, 0, -1.7252812) body0_ke_bulk=8.467297 body0_ke_spin=0.294627 body0_L_bulk_y=0.269524 body0_L_spin_y=0.626767 body1_pos=LPoint3f(13.786854, 25, 13.350294) body1_vel=LVector3f(1.7360403, 0, 1.7252812) body1_omega=-1.930953 body1_momentum=LVector3f(1.7360403, 0, 1.7252812) body1_ke_bulk=2.995216 body1_ke_spin=1.242860 body1_L_bulk_y=-0.609550 body1_L_spin_y=-1.287302 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=11.462512 total_ke_spin=1.537487 total_ke=12.999999 total_L_bulk_y=-0.340027 total_L_spin_y=-0.660535 total_L_y=-1.000562 Comments: Angular Momentum Before: Ly=-1.000000 • Lbulk,y=-3.000000 • Lspin,y=2.000000 After: Ly=-1.000562≈-1.000000 • Lbulk,y=-0.340027

	• $L_{\text{spin},y} = -0.660535$ Conclusion: Total angular momentum is conserved to numerical precision ($\Delta L_y \approx -5.6 \times 10^{-4}$). There is strong redistribution between bulk and spin, and the net spin angular momentum reverses sign overall (from +2.0 to ≈ -0.66), indicating a significant tangential impulse/torque during collision. Energy Before: • $KE_{\text{bulk}} = 10.000000$ • $KE_{\text{spin}} = 3.000000$ • Total $E = 13.000000$ After: • $KE_{\text{bulk}} = 11.462512$ • $KE_{\text{spin}} = 1.537487$ • Total $E = 12.999999 \approx 13$ Conclusion: Total kinetic energy is conserved. Energy is transformed from rotational (spin) kinetic energy into translational kinetic energy (spin KE decreases by ≈ 1.462513 while bulk KE increases by ≈ 1.462512, up to rounding). This also coincides with the post-collision motion gaining a substantial z-component, meaning translational energy is redistributed into additional directions. Conservation Summary • Total linear momentum: conserved; total momentum remains $(-2, 0, 0)$ • Total angular momentum: conserved to numerical precision • Total energy: conserved to numerical precision • Individual components (bulk vs spin energy and angular momentum): not conserved separately, but redistributed during the collision.
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b) Coefficient of Restitution = 0.0

Scenario	Initial and Final Metrics
0	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0)

body0_ke_bulk=2.000000 body0_ke_spin=0.000000
body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000
body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0)
body1_omega=0.000000
body1_momentum=LVector3f(-4, 0, 0)
body1_ke_bulk=8.000000 body1_ke_spin=0.000000
body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000
total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000
total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000

Final:

body0_pos=LPoint3f(-10.751389, 25, -8.251992)
body0_vel=LVector3f(-1.3606727, 0, -1.12008) body0_omega=-1.223571
body0_momentum=LVector3f(-1.3606727, 0, -1.12008)
body0_ke_bulk=1.553005 body0_ke_spin=0.499042
body0_L_bulk_y=-0.814155 body0_L_spin_y=-0.815714
body1_pos=LPoint3f(-4.21521, 25, 8.251992) body1_vel=LVector3f(-0.63932746, 0, 1.12008) body1_omega=-1.223571
body1_momentum=LVector3f(-0.63932746, 0, 1.12008)
body1_ke_bulk=0.831659 body1_ke_spin=0.499042
body1_L_bulk_y=-0.554353 body1_L_spin_y=-0.815714
total_momentum=LVector3f(-2.0000002, 0, 0)
total_ke_bulk=2.384664 total_ke_spin=0.998084 total_ke=3.382748
total_L_bulk_y=-1.368508 total_L_spin_y=-1.631428 total_L_y=-2.999936

Comments:

Angular Momentum

Before:

$L_y = -3.000000$ (all bulk, no spin)

After:

$L_y = -2.999936 \approx -3.000000$

- $L_{\text{bulk},y} = -1.368508$
- $L_{\text{spin},y} = -1.631428$

Conclusion: Total angular momentum is conserved to numerical precision ($\Delta L_y \approx +6.4 \times 10^{-5}$). Even with $e=0$, the collision redistributes angular momentum between bulk and spin (bulk becomes less negative while spin becomes more negative), with the total essentially unchanged.

Energy

Before:

- $KE_{\text{bulk}} = 10.000000$
- $KE_{\text{spin}} = 0.000000$

	• Total E=10.000000 After: • KEbulk=2.384664 • KEspin=0.998084 • Total E=3.382748 Conclusion: Total mechanical energy is not conserved (consistent with e=0). A large amount of translational kinetic energy is removed from the tracked mechanical energy: most of the initial bulk KE is converted into untracked internal energy (deformation/heat/sound), while a smaller portion becomes rotational (spin) kinetic energy: • bulk KE drops by ≈ 7.615336 • spin KE increases by ≈ 0.998084 • net mechanical energy loss ≈ 6.617252 Conservation Summary • Total linear momentum: conserved to numerical precision (total momentum $\approx (-2, 0, 0)$) • Total angular momentum: conserved to numerical precision • Total energy: not conserved (mechanical energy decreases strongly for e=0) • Individual components (bulk vs spin): redistributed; most “lost” energy has been transformed into internal/non-mechanical forms not tracked in the printout.
1	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000 total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000 Final: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000

body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000
body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0)
body1_omega=0.000000
body1_momentum=LVector3f(-4, 0, 0)
body1_ke_bulk=8.000000 body1_ke_spin=0.000000
body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000
total_momentum=LVector3f(-2, 0, 0)
total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000
total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000

Comments:

Angular Momentum

Before:

Ly=-3.000000 (all bulk, no spin)

After:

Ly=-3.000000 (all bulk, no spin)

- Lbulk,y=-3.000000
- Lspin,y=0.000000

Conclusion: Total angular momentum is conserved exactly, and there is **no redistribution** between bulk and spin (nothing changes).

Energy

Before:

- KEbulk=10.000000
- KEs spin=0.000000
- Total E=10.000000

After:

- KEbulk=10.000000
- KEs pin=0.000000
- Total E=10.000000

Conclusion: Total energy is conserved exactly, and **no energy is transformed** (the system state is unchanged).

Conservation Summary

- **Total linear momentum:** conserved exactly
- **Total angular momentum:** conserved exactly
- **Total energy:** conserved exactly
- **Interpretation:** this output indicates **no collision/interaction occurred**

	(or the collision produced zero impulse), since all quantities are identical before and after.
2	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=0.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=0.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=0.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=10.000000 total_ke_spin=0.000000 total_ke=10.000000 total_L_bulk_y=-3.000000 total_L_spin_y=0.000000 total_L_y=-3.000000 Final: body0_pos=LPoint3f(-1.2244186, 25, -9.270607) body0_vel=LVector3f(-0.025479793, 0, -1.1694112) body0_omega=-0.574702 body0_momentum=LVector3f(-0.025479793, 0, -1.1694112) body0_ke_bulk=0.684086 body0_ke_spin=0.110094 body0_L_bulk_y=-1.195636 body0_L_spin_y=-0.383135 body1_pos=LPoint3f(-13.74216, 25, 9.270607) body1_vel=LVector3f(-1.9745202, 0, 1.1694112) body1_omega=1.220149 body1_momentum=LVector3f(-1.9745202, 0, 1.1694112) body1_ke_bulk=2.633126 body1_ke_spin=0.496255 body1_L_bulk_y=-2.234764 body1_L_spin_y=0.813433 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=3.317212 total_ke_spin=0.606349 total_ke=3.923561 total_L_bulk_y=-3.430400 total_L_spin_y=0.430298 total_L_y=-3.000102 Comments: Angular Momentum Before: $L_y = -3.000000$ After: $L_y = -3.000102 \approx -3.000000$ • $L_{bulk,y} = -3.430400$ • $L_{spin,y} = 0.430298$ Conclusion: Total angular momentum is conserved to numerical precision ($\Delta L_y \approx -1.0 \times 10^{-4}$). The collision redistributes angular momentum between bulk

	and spin (bulk becomes more negative while spin becomes positive), with the total essentially unchanged. Energy Before: • KEbulk=10.000000 • KEspin=0.000000 • Total E=10.000000 After: • KEbulk=3.317212 • KEspin=0.606349 • Total E=3.923561 Conclusion: Total mechanical energy is not conserved (consistent with $e=0$). Most of the initial translational kinetic energy is converted into untracked internal energy (deformation/heat/sound), while a smaller portion becomes rotational (spin) kinetic energy: • bulk KE drops by ≈ 6.682788 • spin KE increases by ≈ 0.606349 • net mechanical energy loss ≈ 6.076439 Conservation Summary • Total linear momentum: conserved exactly (total momentum remains $(-2, 0, 0)$) • Total angular momentum: conserved to numerical precision • Total energy: not conserved (mechanical energy decreases strongly for $e=0$) • Individual components (bulk vs spin): redistributed; most “lost” energy has been transformed into internal/non-mechanical forms not tracked in the printout.
3	Initial: <pre>body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=-5.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=8.333333 body0_L_bulk_y=-1.000000 body0_L_spin_y=-3.333333 body1_pos=LPoint3f(-1.9, 25, 1.9) body1_vel=LVector3f(0, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(0, 0, 0) body1_ke_bulk=0.000000 body1_ke_spin=0.000000 body1_L_bulk_y=0.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(2, 0, 0)</pre>

total_ke_bulk=2.000000 total_ke_spin=8.333333 total_ke=10.333333
total_L_bulk_y=-1.000000 total_L_spin_y=-3.333333 total_L_y=-4.333333

Final:

body0_pos=LPoint3f(13.966736, 25, -3.8357033) body0_vel=LVector3f(2, 0, -0.402698) body0_omega=-4.881377
body0_momentum=LVector3f(2, 0, -0.402698)
body0_ke_bulk=2.081083 body0_ke_spin=7.942615
body0_L_bulk_y=-2.047030 body0_L_spin_y=-3.254252
body1_pos=LPoint3f(-1.9, 25, 5.2356796) body1_vel=LVector3f(0, 0, 0.402698)
body1_omega=0.304212
body1_momentum=LVector3f(0, 0, 0.402698)
body1_ke_bulk=0.081083 body1_ke_spin=0.030848
body1_L_bulk_y=0.765126 body1_L_spin_y=0.202808
total_momentum=LVector3f(2, 0, 0)
total_ke_bulk=2.162166 total_ke_spin=7.973463 total_ke=10.135629
total_L_bulk_y=-1.281904 total_L_spin_y=-3.051443 total_L_y=-4.333347

Comments:

Angular Momentum

Before:

Ly=-4.333333

- Lbulk,y=-1.000000
- Lspin,y=-3.333333

After:

Ly=-4.333347≈-4.333333

- Lbulk,y=-1.281904
- Lspin,y=-3.051443

Conclusion: Total angular momentum is conserved to numerical precision ($\Delta Ly \approx -1.4 \times 10^{-5}$). Angular momentum is transferred **from spin to bulk** (spin becomes less negative, bulk becomes more negative), with redistribution between the two bodies.

Energy

Before:

- KEbulk=2.000000
- KEs spin=8.333333
- Total E=10.333333

After:

- KEbulk=2.162166

	• KEspin=7.973463 • Total E=10.135629 Conclusion: Total mechanical energy is not conserved (consistent with e=0). Some initial mechanical energy--primarily from rotational (spin) kinetic energy--is converted into untracked internal energy (deformation/heat/sound), while a small portion is redistributed into translation: • bulk KE increases by ≈ 0.162166 • spin KE decreases by ≈ 0.359870 • net mechanical energy loss ≈ 0.197704 Conservation Summary • Total linear momentum: conserved exactly; total momentum remains (2,0,0) • Total angular momentum: conserved to numerical precision • Total energy: not conserved (mechanical energy decreases for e=0) • Individual components (bulk vs spin): redistributed; the "lost" mechanical energy has been transformed into internal/non-mechanical forms not tracked in the printout.
4	Initial: body0_pos=LPoint3f(-6, 25, -0.5) body0_vel=LVector3f(2, 0, 0) body0_omega=3.000000 body0_momentum=LVector3f(2, 0, 0) body0_ke_bulk=2.000000 body0_ke_spin=3.000000 body0_L_bulk_y=-1.000000 body0_L_spin_y=2.000000 body1_pos=LPoint3f(11, 25, 0.5) body1_vel=LVector3f(-4, 0, 0) body1_omega=0.000000 body1_momentum=LVector3f(-4, 0, 0) body1_ke_bulk=8.000000 body1_ke_spin=0.000000 body1_L_bulk_y=-2.000000 body1_L_spin_y=0.000000 total_momentum=LVector3f(-2, 0, 0) total_ke_bulk=10.000000 total_ke_spin=3.000000 total_ke=13.000000 total_L_bulk_y=-3.000000 total_L_spin_y=2.000000 total_L_y=-1.000000 Final: body0_pos=LPoint3f(-12.20841, 25, -7.629465) body0_vel=LVector3f(-1.5169648, 0, -0.9577374) body0_omega=2.792436 body0_momentum=LVector3f(-1.5169648, 0, -0.9577374) body0_ke_bulk=1.609222 body0_ke_spin=2.599233 body0_L_bulk_y=-0.118820 body0_L_spin_y=1.861624 body1_pos=LPoint3f(-2.758302, 25, 7.629465) body1_vel=LVector3f(-0.48303496, 0, 0.9577374) body1_omega=-2.548957 body1_momentum=LVector3f(-0.48303496, 0, 0.9577374) body1_ke_bulk=0.575292 body1_ke_spin=2.165727 body1_L_bulk_y=-1.043570 body1_L_spin_y=-1.699305 total_momentum=LVector3f(-1.9999998, 0, 0)

total_ke_bulk=2.184513 total_ke_spin=4.764961 total_ke=6.949474
total_L_bulk_y=-1.162390 total_L_spin_y=0.162320 total_L_y=-1.000070

Comments:

Angular Momentum

Before:

$L_y = -1.000000$

- $L_{\text{bulk},y} = -3.000000$
- $L_{\text{spin},y} = 2.000000$

After:

$L_y = -1.000070 \approx -1.000000$

- $L_{\text{bulk},y} = -1.162390$
- $L_{\text{spin},y} = 0.162320$

Conclusion: Total angular momentum is conserved to numerical precision ($\Delta L_y \approx -7.0 \times 10^{-5}$). The collision redistributes angular momentum between bulk and spin; net spin angular momentum decreases strongly (from +2.0 to $\approx +0.16$).

Energy

Before:

- $KE_{\text{bulk}} = 10.000000$
- $KE_{\text{spin}} = 3.000000$
- Total E = 13.000000

After:

- $KE_{\text{bulk}} = 2.184513$
- $KE_{\text{spin}} = 4.764961$
- Total E = 6.949474

Conclusion: Total mechanical energy is **not** conserved (consistent with $e=0$). A large fraction of the initial **translational kinetic energy** is converted into **untracked internal energy** (deformation/heat/sound). Some energy is also redistributed into spin:

- bulk KE decreases by ≈ 7.815487
- spin KE increases by ≈ 1.764961
- net mechanical energy loss ≈ 6.050526

Conservation Summary

- **Total linear momentum:** conserved to numerical precision (total momentum $\approx (-2, 0, 0)$)

- | | |
|--|--|
| | <ul style="list-style-type: none">• Total angular momentum: conserved to numerical precision• Total energy: not conserved (mechanical energy decreases strongly for $e=0$)• Individual components (bulk vs spin): redistributed; most “lost” energy has been transformed into internal/non-mechanical forms not tracked in the printout. |
|--|--|